

Artificial Intelligence in Healthcare Management and Economics

Evidence, Value, Governance, and Lifecycle Decision-Making

START HERE

Digital Resource Companion

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Purpose of the companion

The Digital Resource Companion is the applied extension of Artificial Intelligence in Healthcare Management and Economics: Evidence, Value, Governance, and Lifecycle Decision-Making. The book contains the durable conceptual, economic, methodological, and governance argument. The companion provides editable institutional records, structured workbooks, worked examples, decision pathways, reference guides, process maps, and reproducible analytical resources that support application and local adaptation.

The companion does not replace the reasoning developed in the book. It provides a structured way to apply that reasoning to a clearly specified jurisdiction, organization, population, workflow, intervention, comparator, technology version, evidence base, and accountable decision process.

Choose the appropriate entry point

Entry point	Best suited to	What it provides
GitHub Pages Web Companion	Most users beginning online	Guided navigation by chapter, decision stage, form, workbook, worked example, or analytical method.
Reader and Practitioner Edition	Readers, practitioners, managers, analysts, educators, and institutional teams	A complete, accessible offline collection with the Web Companion, forms, workbooks, cases, guides, selected analytics, catalogues, and printable companion.
Core Tools Pack	Teams that primarily need the principal institutional documents	The consolidated Core Resource Forms, consolidated Core Resource Workbook, practical guidance, licence, disclaimer, and package manifest.
Canonical Source	Researchers, developers, repository maintainers, advanced analysts, educators, and institutions	The complete authoritative source collection, including editable, computational, machine-readable, process, publication, and website materials.
Printable Digital Resource Companion	Readers who prefer a coherent readable reference	A book-like PDF and Word volume explaining the companion architecture, resources, cases, methods, and recommended use.

Most readers should begin with the Web Companion or the Reader and Practitioner Edition. The Core Tools Pack is a convenience subset for focused work, not a separate edition. The Canonical Source supports maintenance, reproduction, technical extension, and preparation of future releases.

Resource scope

Resource class	V1.0 scope
Book structure	4 parts and 12 chapters
Principal book figures	29
Book tables	38, including the two front-matter tables
Core institutional records	19
Worked examples	16, plus AIHE-CASE00 as the common case-analysis method
Python analytical modules	36
Catalogued companion resources	126

The four core resource families

Family	Scope	Core identifiers
A — Framing and readiness	Decision specification, participation, readiness, and data fitness.	AIHE-A01 to AIHE-A04
B — Economics and modelling	Economic specification, lifecycle cost, benefits, HTA, uncertainty, and reproducibility.	AIHE-B01 to AIHE-B05
C — Quality and institutional risk	Quality control, corrective action, risk, human oversight, and communication.	AIHE-C01 to AIHE-C05
D — Procurement and lifecycle governance	Supplier requirements, current decision states, monitoring, incidents, material change, renewal, replacement, and retirement.	AIHE-D01 to AIHE-D05

Stable resource identifiers remain unchanged for retained resources. Examples include AIHE-D02 for current decision-state documentation, AIHE-CASE12 for a model-update and renewal case, and AIHE-PY40 for advisory decision-state and trigger review.

Recommended institutional sequence

1. Define the service problem, affected population, setting, intended use, accountable decision-maker, and realistic comparator.
2. Read the aligned book chapter and identify the smallest sufficient companion resource set.
3. Review the corresponding worked example and distinguish its synthetic assumptions from evidence needed locally.
4. Create a governed local copy of each selected record or workbook.
5. Replace illustrative content with authorized, versioned local evidence and preserve important unknowns or disagreements.
6. Complete multidisciplinary review proportionate to consequence, uncertainty, scale, reversibility, and institutional dependence.
7. Record the current decision state, conditions, mandatory safeguards, monitoring thresholds, transition triggers, accountable authority, next review date, and credible exit route.
8. Reconsider the decision after material change, new evidence, an incident, changed costs, renewal, restriction, replacement, or retirement.

Using forms and workbooks together

Word forms	Excel workbooks
Narrative justification, committee discussion, conditions, responsibilities, dissent, authorization, and sign-off.	Structured data entry, costing, registers, comparisons, calculations, monitoring, and repeated review.
Use the consolidated volume when several institutional records are needed; choose an individual form for a focused task.	Use the consolidated workbook for several records; choose a specialist workbook or analytical template for a focused calculation.
Preserve the identifier, version, evidence cut-off, intervention identity, owner, and local changes.	Preserve the same controls, with data definitions, formula assumptions, input sources, and calculation checks.

A form and workbook can be used together. The form preserves the narrative decision; the workbook holds structured supporting evidence. Neither replaces the current decision dossier.

Decision lifecycle at a glance

The sequence is iterative. Later evidence or changed circumstances may reopen an earlier decision. Implementation activity remains distinct from institutional permission.

Stage	Principal question	Typical core records
1. Define the decision	Is the problem sufficiently important and specified to justify further assessment?	AIHE-A01; AIHE-A02
2. Assess readiness and data	Are critical prerequisites adequate for the intended level of consequence and dependence?	AIHE-A03; AIHE-A04
3. Establish evidence and quality requirements	Is the planned evidence proportionate to intended use, uncertainty, and consequence?	AIHE-C01; AIHE-C02; AIHE-EX05; AIHE-RG02; AIHE-RG06
4. Model cost, value, affordability, and HTA	Does the intervention have a defensible value proposition and a financeable path under realistic scenarios?	AIHE-B01; AIHE-B02; AIHE-B03; AIHE-B04
5. Select the decision model and characterize uncertainty	Is model complexity justified, credible, reproducible, and decision-relevant?	AIHE-B05; AIHE-RG05
6. Assess governance and institutional risk	Are mandatory safeguards satisfied and are residual risks explicitly owned and governable?	AIHE-C03; AIHE-C04; AIHE-C05; AIHE-RG04
7. Procure and contract	Can the institution enforce the evidence, monitoring, change, continuity, and exit conditions required for safe use?	AIHE-D01
8. Implement, monitor, and control change	Does real-world use remain within approved performance, safety, workload, equity, cost, and governance conditions?	AIHE-D02; AIHE-D03; AIHE-D04; AIHE-D05; AIHE-RG03
9. Reassess continuing value, affordability, and support	Does continuing use remain justified, affordable, supportable, and preferable to current alternatives without compromising mandatory safeguards?	AIHE-B02; AIHE-B03; AIHE-B04; AIHE-D01; AIHE-D05
10. Assemble the decision record and renew or retire	Is continuing use still preferable to redesign, restriction, replacement, or retirement under current evidence and alternatives?	AIHE-D02; AIHE-D05; AIHE-P; AIHE-AE

Worked examples and computational resources

The sixteen worked examples and AIHE-CASE00 common method connect service problems with evidence, economics, governance, implementation, monitoring, and lifecycle decisions. Their scenarios and demonstration datasets are synthetic unless explicitly stated; they are not empirical evidence for a local decision.

The Canonical Source contains thirty-six Python analytical modules. Supplementary worked cases address financial recognition and budget incidence, the expected value of additional evidence, and upstream model-provider change. Method choice, versioned inputs, verification, validation, and expert interpretation remain necessary.

Version control and responsible use

- Record the stable identifier, exact resource version, release date, access date, evidence cut-off, intervention version, and local modifications.
- Preserve the exact data, code, environment, assumptions, and outputs used in consequential analysis. Verify the currency of law, standards, software, and supplier information before reliance.
- A completed form, score, or model result does not authorize use. Mandatory safeguards remain non-compensatory, and consequential transitions require accountable human authority.
- Do not enter patient identifiers, credentials, private keys, confidential contracts, or restricted institutional information into public demonstrations or uncontrolled files.
- Maintain a credible route for restriction, rollback, replacement, retirement, and responsible closure.

The companion is an author-developed educational and institutional decision-support collection. It does not replace clinical, legal, regulatory, ethical, privacy, cybersecurity, procurement, accreditation, accounting, financial, or health technology assessment review.

Access and citation

Resource	Address
Persistent companion record (Concept DOI identified in the book)	https://doi.org/10.5281/zenodo.21753927
Canonical repository recorded in the source package	https://github.com/ai-healthcare-economics/digital-resource-companion
Web Companion recorded in the source package	https://ai-healthcare-economics.github.io/digital-resource-companion/

This package is V1.0. The persistent record is not a newly assigned version-specific DOI. When the package is published, cite the version-specific DOI shown on that archived release and retain the exact resource identifier and local changes.